

Benjamin J. Brown

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Education

- 2022–Present **Ph.D. in Physics**, *Brown University*
Advisor: Prof. Gang Xiao
Researching magnetic tunnel junctions for ultrasensitive applications.
- 2022–2024 **Sc.M. in Physics**, *Brown University*
- 2018–2022 **B.Sc. in Physics**, *University of Maryland, Baltimore County*
- 2018–2022 **B.Sc. in Computer Science**, *University of Maryland, Baltimore County*

Awards

- 2025 IOP Publishing Outstanding Reviewer Award
- 2025 Forrest Award for Excellent Work Related to Experimental Apparatus
- 2020 The Donald N. Langenberg Undergraduate Research Award in Physics

Publications

- [1] CM Lavelle, W Shanks, C Chiang, M Nichols, J Osborne, A Herschelman, Benjamin J. Brown and M Cho, “Approaches for single channel large area silicon photomultiplier array readout”, *AIP Advances* **9**:3 (2019). doi:10.1063/1.5088503.
- [2] Vineetha S. Bheemarasetty, David T. Plouff, Xinhao Wang, Benjamin J. Brown, H Minh Cao, John Q Xiao and Gang Xiao, “Shubnikov–de Haas oscillations in half-metallic ferromagnetic CrO₂ epitaxial films”, *Physical Review B* **111**:21, p. 214440 (2025). doi:10.1103/v73x-5ptd.
- [3] Benjamin J. Brown, Liam K Mitchell, Vineetha S Bheemarasetty, H Minh Cao, Justin N Kingsnorth, Jerome N Sanes and Gang Xiao, “High magnetic sensitivity via large-diameter-vortex stability in magnetic tunnel junctions through controlled anisotropy”, *Physical Review Applied* **24**:3, p. 034063 (2025). doi:10.1103/41by-5p3c.
- [4] H. Minh Cao, Benjamin J. Brown, Vineetha S. Bheemarasetty and Gang Xiao, “Annealing and Thickness Optimisation in CoZrNb and FeSiBNbCu Thin Films for Low-Frequency Magnetic Flux Concentrators”, *Journal of Magnetism and Magnetic Materials* (2025). doi:0.1016/j.jmmm.2025.173621.
- [5] Z. C. Adamson, Liam K. Mitchell, Benjamin J. Brown, William R Patterson, Gang Xiao and A Zaslavsky, “Integration of MTJ devices with cryo-CMOS readout circuitry for magnetic field sensing”, *Solid State Electronics*, p. 109312 (2025). doi:0.1016/j.sse.2025.109312.
- [6] Liam K. Mitchell, Benjamin J. Brown, Z C Adamson, A Zaslavsky and Gang Xiao, “Contrasting uniaxial and vortex-state magnetic tunnel junction sensors: sensitivity and tunnel magnetoresistance at low temperatures”, *Journal of Physics D: Applied Physics* **59**:5, p. 055004 (2026). doi:10.1088/1361-6463/ae3df2.
- [7] Benjamin J. Brown, Liam K Mitchell, Hongzhou Yu and Gang Xiao, “Reassessing magnetic tunnel junction detectability for ultrasensitive sensing using small-field sensitivity and Jiles–Atherton modeling”, *Journal of Physics D: Applied Physics* **59**:26, p. 265002 (2026). doi:10.1088/1361-6463/ae802f.

Under Review

- [1] Liam K Mitchell, Benjamin J Brown and Gang Xiao. “Electrical imaging of nanoscale pinning landscapes using magnetic vortex cores”. Submitted to Nature Communications. 2026.

- [2] Benjamin J Brown, Liam K Mitchell, Haoyu Wu and Gang Xiao. “Scaling laws governing intrinsic detectability in vortex magnetic tunnel junctions”. Submitted to nps Spintronics. 2026.
- [3] Haoyu Wu, Benjamin J Brown and Gang Xiao. “Intermittent Lorentzian noise of magnetic tunnel junctions under voltage bias stress”. Submitted to the Journal of Applied Physics. 2026.

Conferences & Talks

- [1] Vineetha Bheemarasetty, Benjamin J. Brown, Liam K. Mitchell, H. Wu, M. Cao and Gang Xiao. “Enhancing Quantum Sensing Devices with High Spin Polarization”. In: *Northeast Quantum Forum*. Poster Presentation. 2024.
- [2] Benjamin J. Brown, H. M. Cao, Liam K. Mitchell, Vineetha Bheemarasetty, J. Sanes and Gang Xiao. “Optimizing Vortex Magnetic Tunnel Junctions for Advanced Magnetic Sensing”. In: *American Physical Society Global Physics Summit* (Anaheim, CA). Oral Presentation. 2025.
- [3] Vineetha Bheemarasetty, Gang Xiao, Benjamin J. Brown and M. Cao. “Quantum Oscillations in Half Metallic Ferromagnetic CrO₂ Epitaxial Films”. In: *American Physical Society Global Physics Summit* (Anaheim, CA). Oral Presentation. 2025.
- [4] H. M. Cao, Benjamin J. Brown, Vineetha Bheemarasetty and Gang Xiao. “Optimising Magnetic Flux Concentration with CoZrNb and FeSiBNbCu Thin Films: Effects of Annealing on Magnetic Properties”. In: *American Physical Society Global Physics Summit* (Anaheim, CA). Poster Presentation. 2025.
- [5] Benjamin J. Brown, Liam K. Mitchell and Gang Xiao. “From Hysteresis to Linearity: Intrinsic Low-Field Sensitivity of Magnetic Tunnel Junctions”. In: *APS Global Physics Summit* (Denver, CO). Oral Presentation. 2026.
- [6] Liam K. Mitchell, Benjamin J. Brown, Z. Adamson, A. Zaslavsky and Gang Xiao. “Uniaxial and vortex-state magnetic tunnel junction sensors for cryogenic quantum applications”. In: *APS Global Physics Summit* (Denver, CO). Oral Presentation. 2026.
- [7] Haoyu Wu, Benjamin J. Brown, Liam K. Mitchell and Gang Xiao. “Vortex-State Dynamics and Noise Scaling in Magnetic Tunnel Junctions for Low-Field Spintronic Applications”. In: *APS Global Physics Summit* (Denver, CO). Oral Presentation. 2026.
- [8] Liam K. Mitchell, Benjamin J. Brown, Z. Adamson, A. Zaslavsky and Gang Xiao. “Magnetic tunnel junction sensors for vortex tracking”. In: *Foundations of Superconducting Digital Logic Program Review* (College Park, MD). Poster Presentation. June 2026.

Patents

- [1] Gang Xiao, Benjamin J. Brown and Liam K. Mitchell. *Electrical Magnetometry Using Magnetic Tunnel Junction Sensors for Quantitative Characterization of Magnetic Hysteresis and Anhysteresis*. U.S. Patent Application No. 63/919,956. Filed November 18, 2025. Nov. 2025.
- [2] Gang Xiao, Liam K. Mitchell and Benjamin J. Brown. *Embedded topological spin probes for electrical imaging of nanoscale energy landscapes*. U.S. Patent Application No. 63/992,223. Filed February 27, 2026. Feb. 2026.
- [3] Gang Xiao and Benjamin J. Brown. *Vortex Magnetic Tunnel Junction Devices, Systems, and Methods for Engineering Intrinsic Magnetic Performance*. U.S. Patent Application No. 64/115,177. Filed July 20, 2026. July 2026.